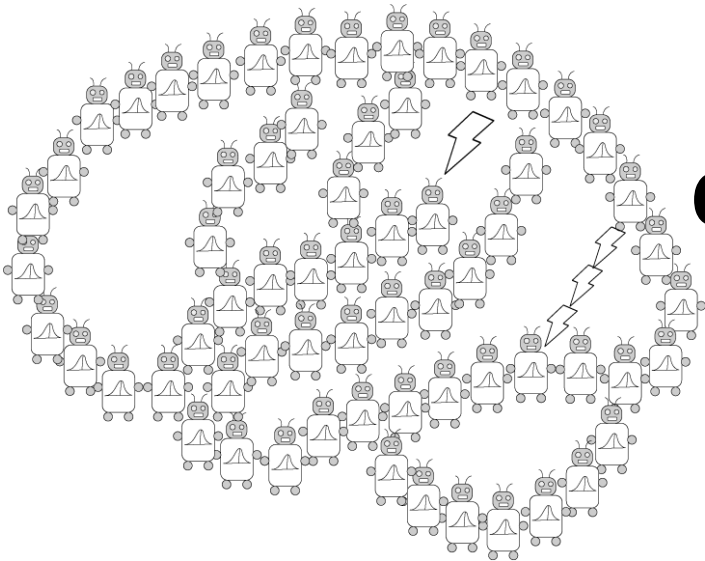


Hierarchical Bayesian Models and Bayesian Networks



**PSYCH/COMP SCI/BMI 841:
Computational Cognitive Science**

Joseph Austerweil

February 20, 2025

- Admin Matters
- Hierarchical Bayesian Modeling
- Bayesian Networks
- **Short rest 1**
- Student presentations:
 - Jess presenting Kemp et al. (2007)
 - Neha presenting Zhu et al. (2007)

Administrative Matters

- Clusters Assignment was due Feb 19 8:59pm
- Generalization Assignment due Mar 5 8:59pm
 - Remember you have three “free” late days for assignments.
 - Late days do not prorate (used up in 24-hour chunks).
- Project proposal due Mar 12 8:59pm
 - No late days for project proposal (or final paper/presentation)
- Questions?

- Admin Matters
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- Bayesian Networks
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Hierarchical Bayesian Framework

(Kemp & Tenenbaum)

$P(\text{form})$

F : form

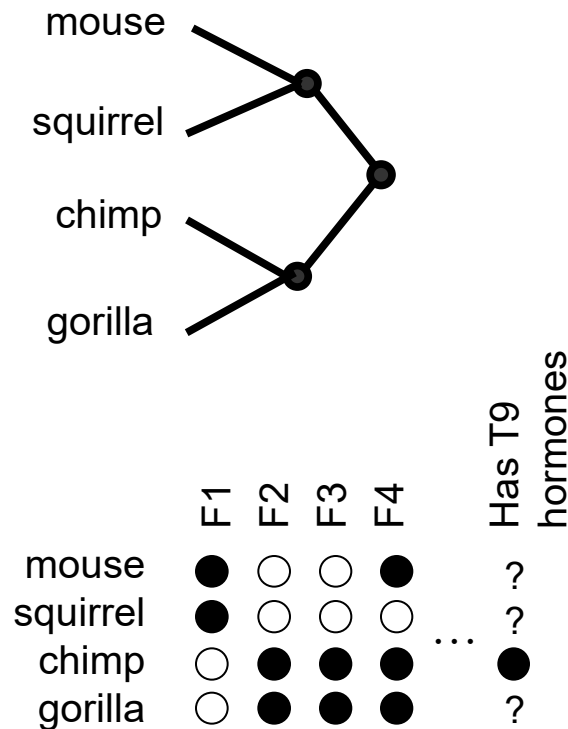
Tree with species at leaf nodes

$\downarrow P(\text{structure} \mid \text{form})$

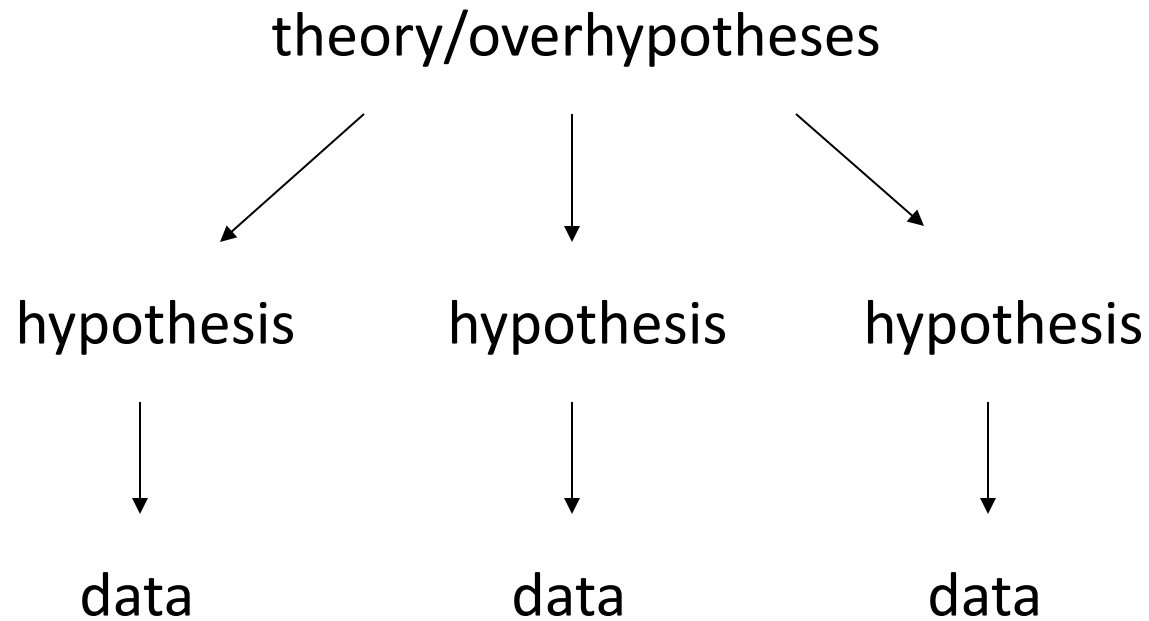
S : structure

$\downarrow P(\text{data} \mid \text{structure})$

D : data



Hierarchical Bayes



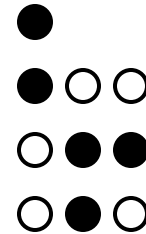
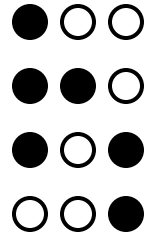
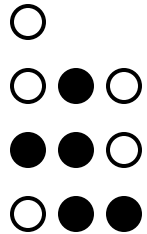
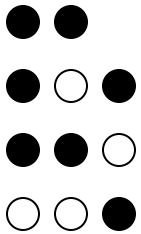
Feature variability (Goodman, 1955)



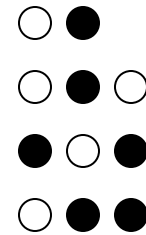
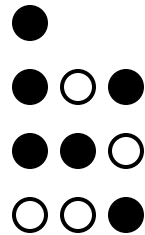
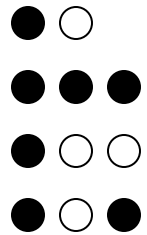
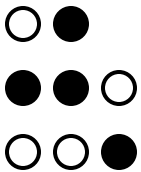
● ?



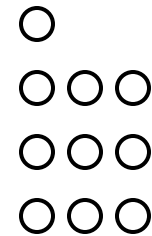
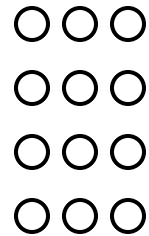
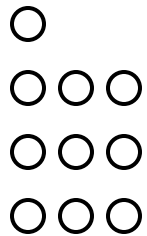
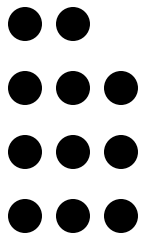
Feature variability (Goodman, 1955)



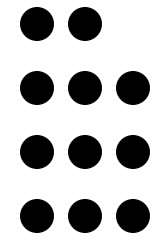
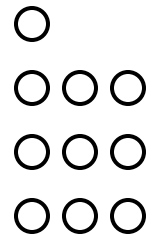
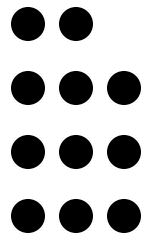
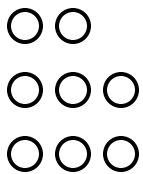
● ?



Feature variability (Goodman, 1955)



● ?



A hierarchical Bayesian model

Overhypothesis

Level 2: Bags in general

most bags are homogeneous

Level 1: Bag proportions

mostly
white

mostly
black

mostly
white

mostly ...
black

Data

○ ○
○ ○
○ ○

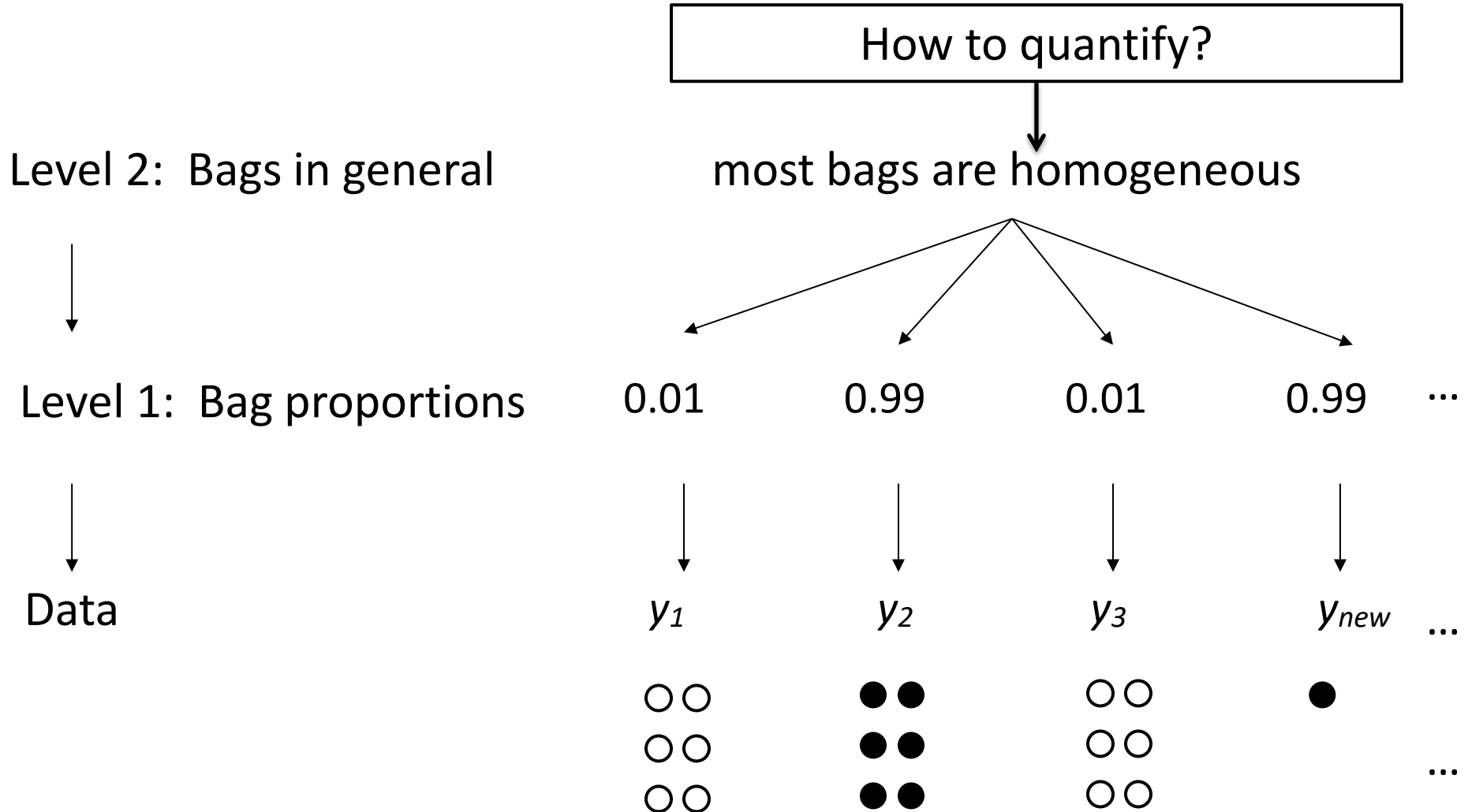
● ●
● ●
● ●

○ ○
○ ○
○ ○

●

...

A hierarchical Bayesian model



Distributions on bag proportions

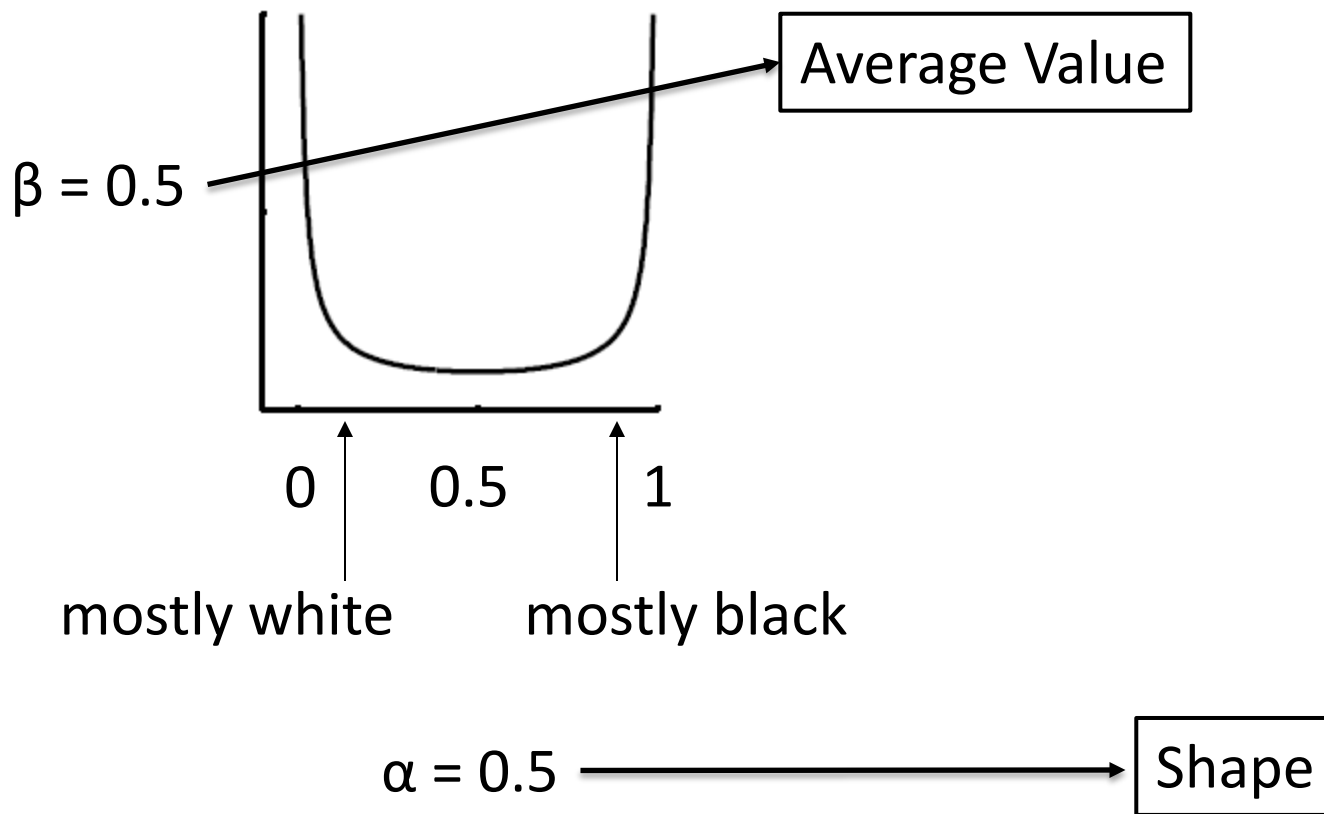
Beta distribution?

$$p(\theta|a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1}$$

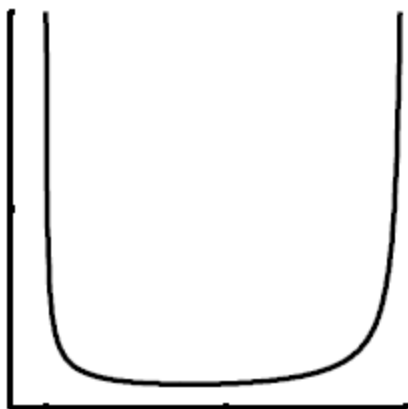
Beta(a, b)

These are Beta($\alpha\beta$, $\alpha(1-\beta)$).

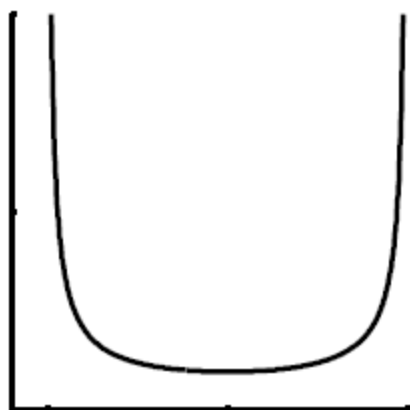
$$p(\theta|\alpha\beta, \alpha(1-\beta)) = \frac{\Gamma(\alpha)}{\Gamma(\alpha\beta)\Gamma(\alpha(1-\beta))} \theta^{\alpha\beta-1} (1-\theta)^{\alpha(1-\beta)-1}$$



$\beta = 0.8$



$\beta = 0.5$



0

0.5

1

mostly white

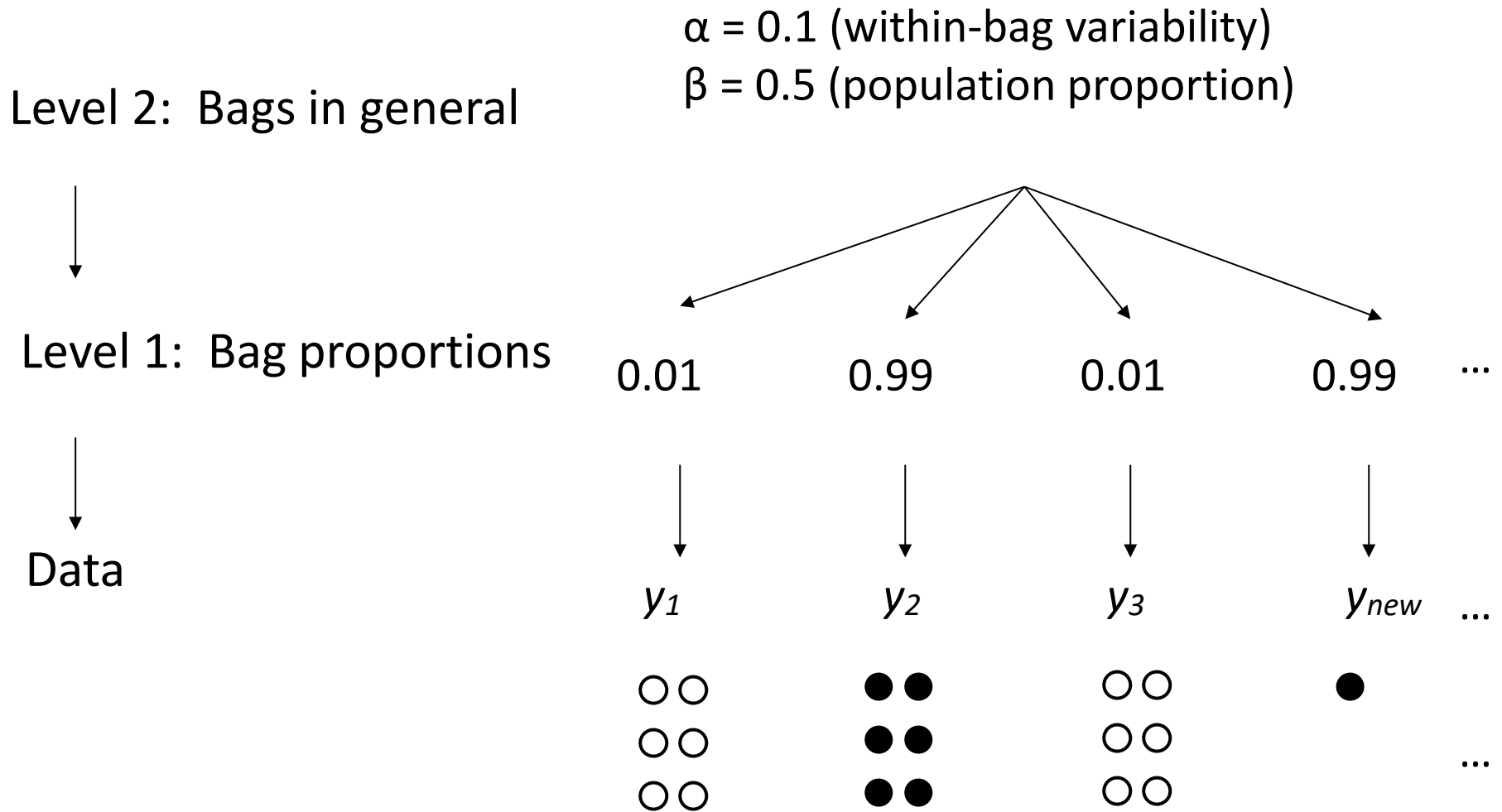
mostly black

$\alpha = 0.5$

$\alpha = 2$

$\alpha = 10$

A hierarchical Bayesian model



A hierarchical Bayesian model

$$\theta_n | \alpha, \beta \stackrel{iid}{\sim} \text{Beta}(\alpha\beta, \alpha(1 - \beta))$$

$$y_n | \theta_n \sim \text{Binomial}(\theta_n, M)$$

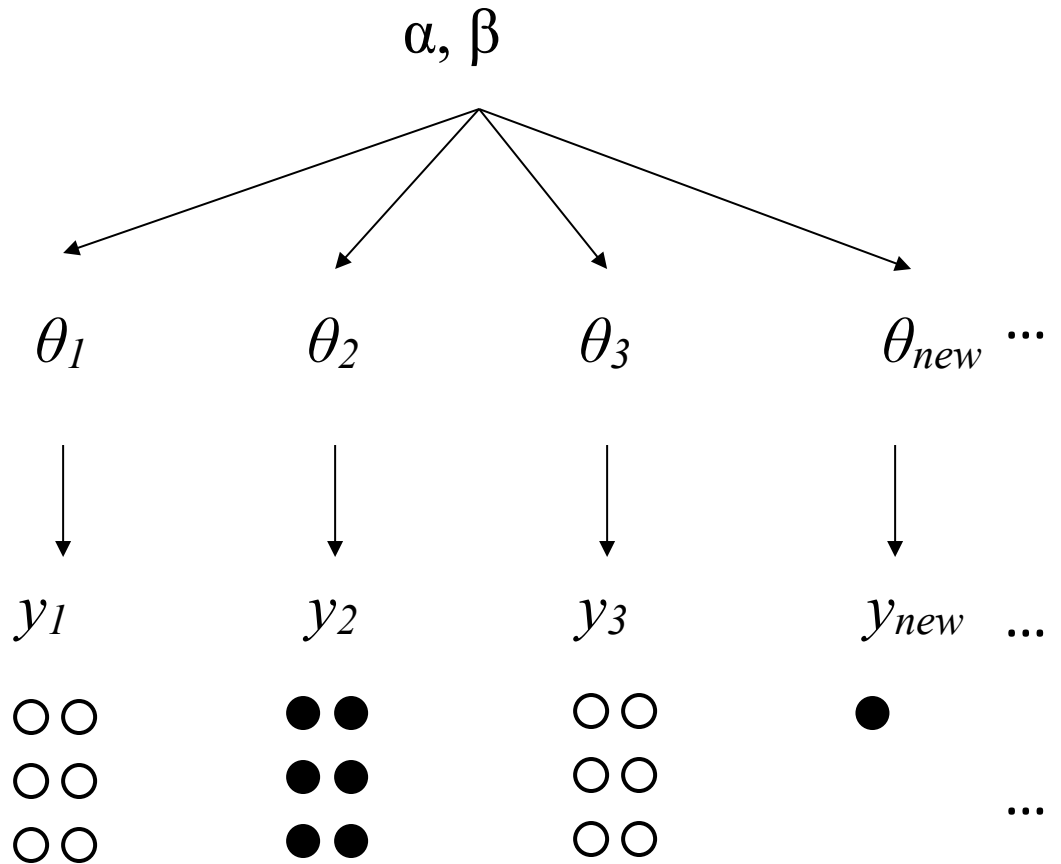
Level 2: Bags in general



Level 1: Bag proportions



Data



Learning

We simultaneously learn α , β , and the θ_i : $p(\alpha, \beta, \theta_1, \theta_2, \dots | y_1, y_2, \dots)$

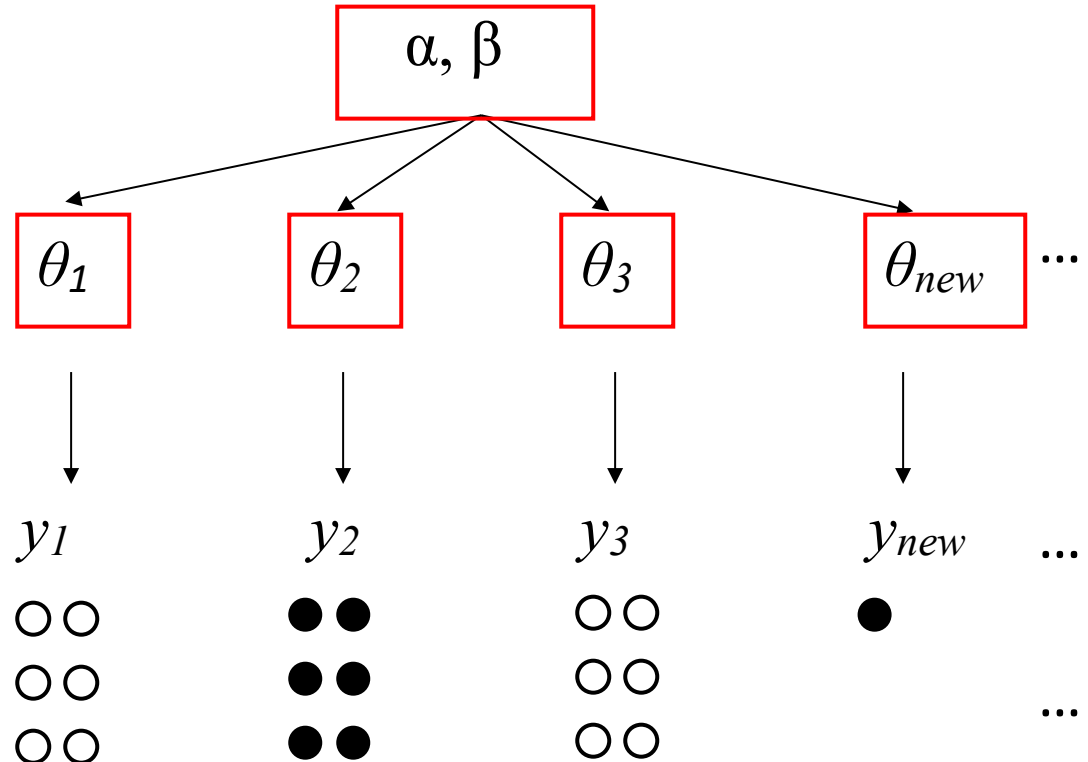
Level 2: Bags in general



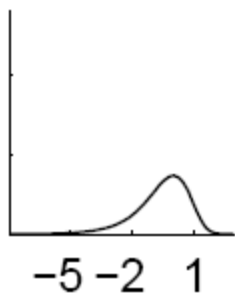
Level 1: Bag proportions



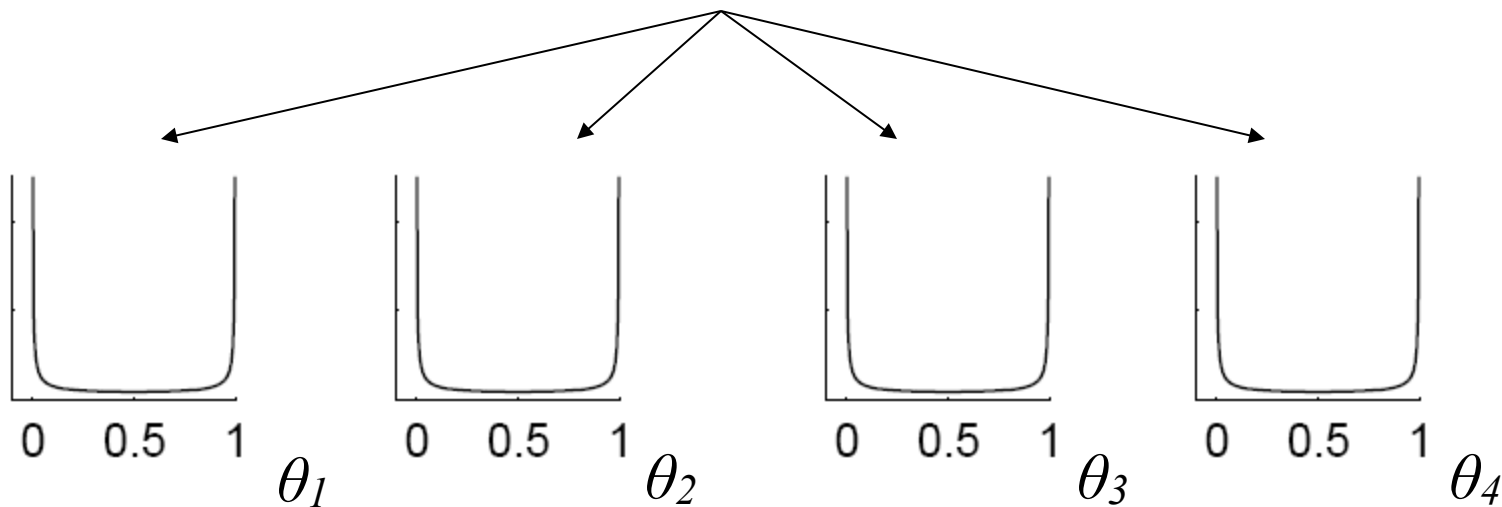
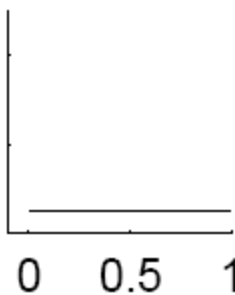
Data



$\log(\alpha)$ (within-bag variability)

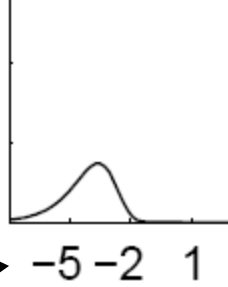


β (population proportion)

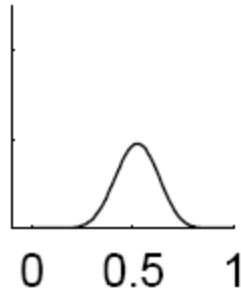


$\log(\alpha)$ (within-bag variability)

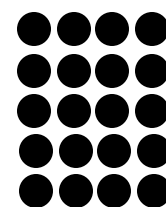
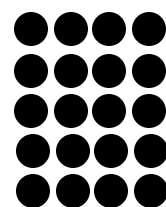
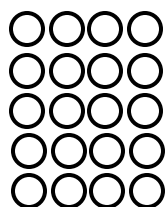
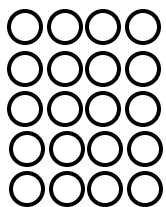
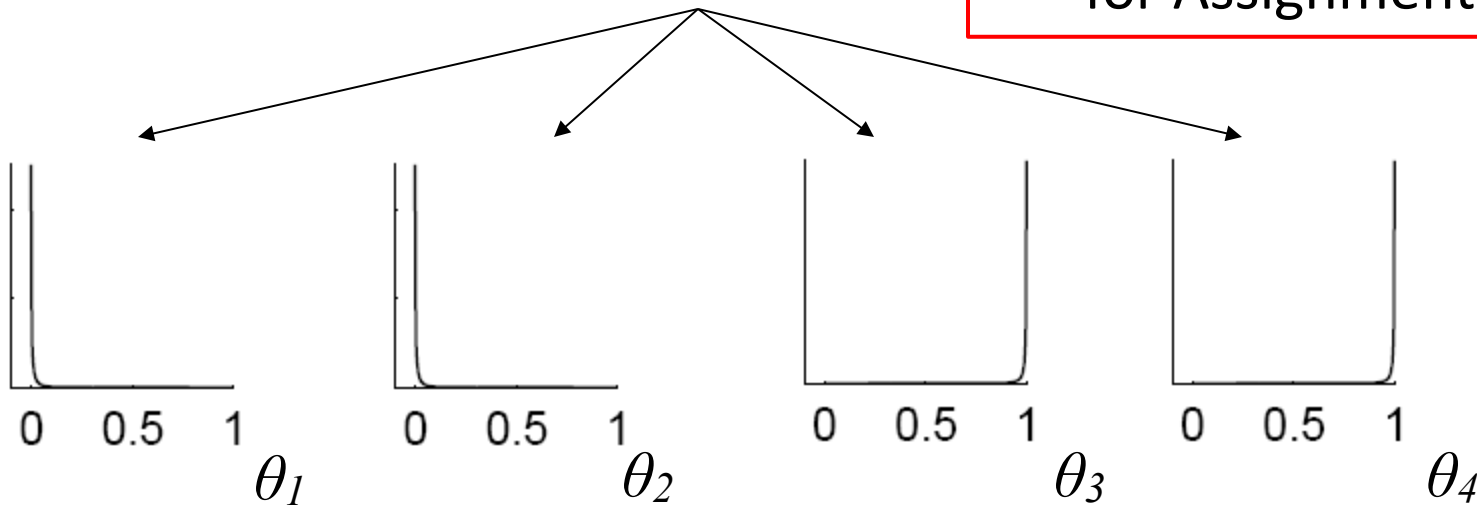
Remember:
negative logs are
very tiny #s!



β (population proportion)



You will implement this
for Assignment 3!



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Challenges of probabilistic models

Specifying well-defined probabilistic models with many variables is hard (for modelers).

Representing probability distributions over those variables is hard (for computers/learners).

Computing quantities using those distributions is hard (for computers/learners).

Challenges of probabilistic models

Specifying well-defined probabilistic models with many variables is hard (for modelers).

Representing probability distributions over those variables is hard (for computers/learners).

Computing quantities using those distributions is hard (for computers/learners).

How do we use fewer numbers?

One random variable:	Domain
X_1 coin toss produces heads	$\{0,1\}$

If I know probability of heads, can I have any value for the probability of tails?

No, it must be $1 - \text{probability of heads}$.

How do we use fewer numbers?

Two random variables:	Domain
-----------------------	--------

X_1 coin toss produces heads	$\{0,1\}$
--------------------------------	-----------

X_4 friend has two-headed coin	$\{0,1\}$
----------------------------------	-----------

Friend is equally likely to have normal or fixed coin.

$$P(X_1=H, X_4=F) = 0.25$$

$$P(X_1=T, X_4=F) = 0.25$$

How do we use fewer numbers?

Two random variables:

Domain

X_1 coin toss produces heads $\{0,1\}$

X_4 friend has two-headed coin $\{0,1\}$

Friend is equally likely to have normal or fixed coin.

$$P(X_1=H, X_4=F) = 0.25 \quad P(X_1=H, X_4=T) = 0.5$$

$$P(X_1=T, X_4=F) = 0.25 \quad P(X_1=T, X_4=T) = 0.0$$

How do we use fewer numbers?

Four random variables:	Domain
X_1 coin toss produces heads	{0,1}
X_2 pencil levitates	{0,1}
X_3 friend has psychic powers	{0,1}
X_4 friend has two-headed coin	{0,1}

Joint Probability Distributions

Requires 15 numbers to specify
probability of all values x_1, x_2, x_3, x_4

– N binary variables, $2^N - 1$ numbers

Statistical independence

Two random variables X_1 and X_2 are *independent* if

$$P(x_1 | x_2) = P(x_1)$$

– e.g. coinflips: $P(x_1 = H | x_2 = H) = P(x_1 = H) = 0.5$

Independence makes it easier to represent and work with probability distributions

We can exploit the product rule:

$$P(x_1, x_2, x_3, x_4) = P(x_1 | x_2, x_3, x_4) P(x_2 | x_3, x_4) P(x_3 | x_4) P(x_4)$$

$$P(x_1, A) = P(x_1 | A) P(A) = P(x_1 | x_2, x_3, x_4) P(x_2, x_3, x_4)$$

$$P(x_1, x_2, B) = P(x_1 | x_2, x_3, x_4) P(x_2 | B) P(B) = P(x_1 | x_2, x_3, x_4) P(x_2 | x_3, x_4) P(x_3, x_4)$$

Statistical independence

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If x_1 , x_2 , x_3 , and x_4 are all independent...

$$P(x_1, x_2, x_3, x_4) = P(x_1)P(x_2)P(x_3)P(x_4)$$

Expressing independence

Statistical independence is the key to efficient probabilistic representation and computation.

This has led to the development of languages for indicating dependencies among variables.

Some of the most popular languages are based on “graphical models.”

Directed graphical models

Express the probabilistic dependencies among a set of variables as a graph (Pearl)

Consist of

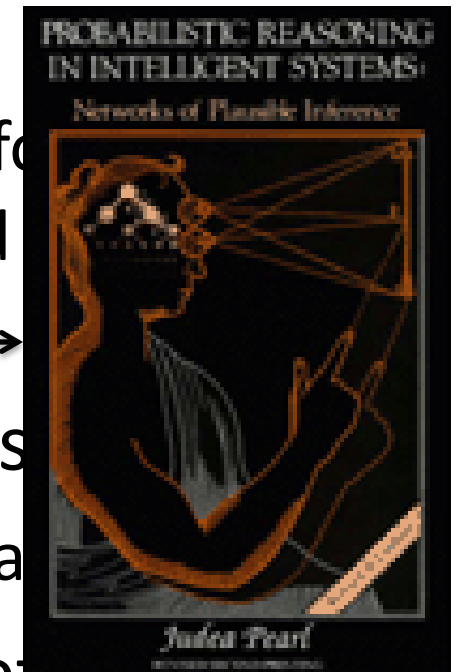
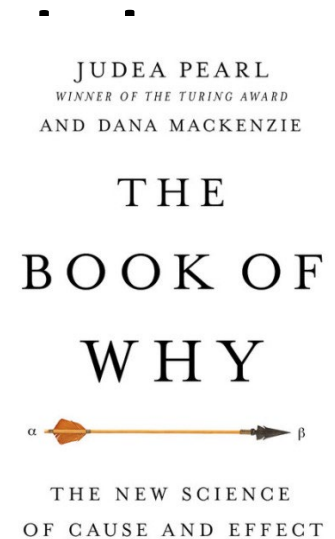
- a set of nodes
- a set of edges
- a *conditional probability distribution* for each node, conditioned on its parents, multiplied together to yield the distribution over variables

Constrained to directed acyclic graphs

- Acyclic just means that no cycles are allowed

Called Bayesian networks

Rumelhart prize winner



The Markov assumption

Every node is conditionally independent of its non-descendants, given its parents

$$P(x_i \mid x_{i+1}, \dots, x_k) = P(x_i \mid \mathbf{Pa}(X_i))$$

where $\mathbf{Pa}(X_i)$ is the set of parents of X_i

$$P(x_1, \dots, x_k) = \prod_{i=1}^k P(x_i \mid \mathbf{Pa}(X_i))$$

(via the product rule)

The two representations are equivalent via Hammerly-Clifford's Theorem...

Efficient representation and inference

Four random variables:

X_1 coin toss produces heads

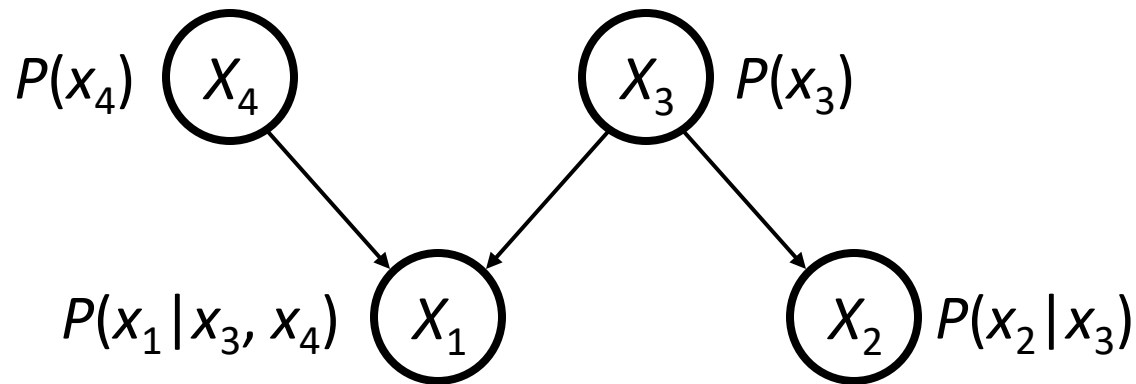
X_2 pencil levitates

X_3 friend has psychic powers

X_4 friend has two-headed coin

Parents of X_1 ?

Parents of X_4 ?



Efficient representation and inference

Four random variables:

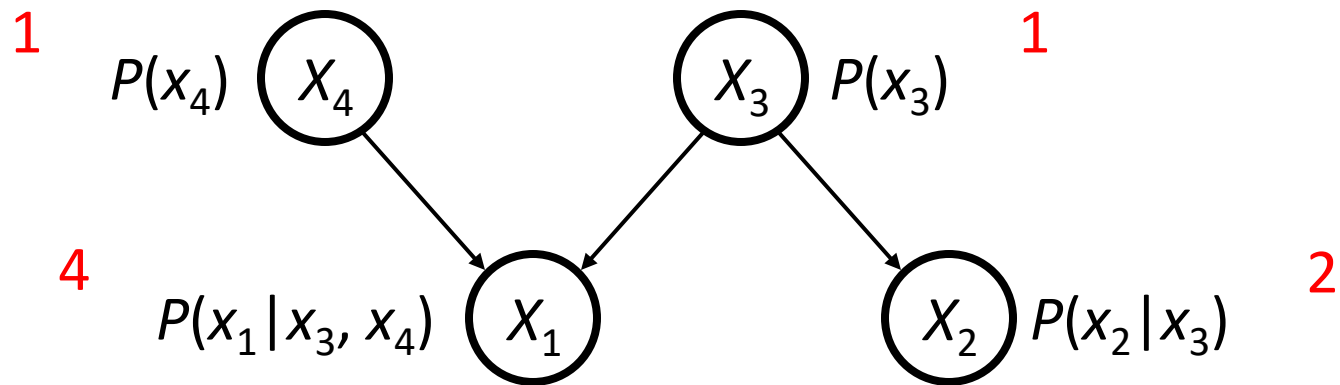
X_1 coin toss produces heads

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How many values
to specify?



total = 8 (vs 15) $P(X_1, X_2, X_3, X_4) = P(X_1 | X_3, X_4) P(X_2 | X_3) P(X_3) P(X_4)$

Reading a Bayesian network

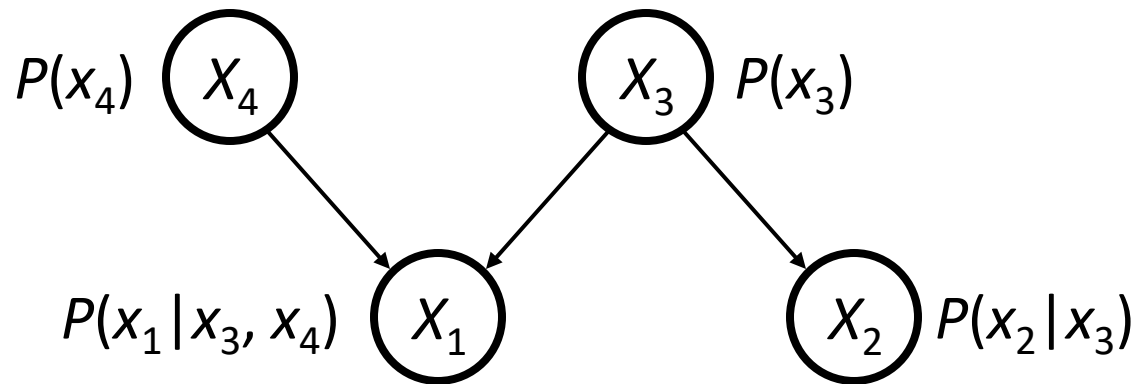
Four random variables:

X_1 coin toss produces heads

X_2 pencil levitates

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Reading a Bayesian network

The structure of a Bayes net can be read as the *generative process* behind a distribution.

Gives the joint probability distribution over variables obtained by sampling each variable conditioned on its parents.

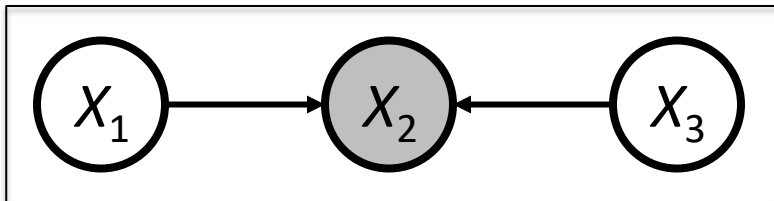
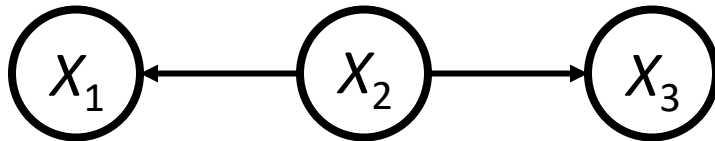
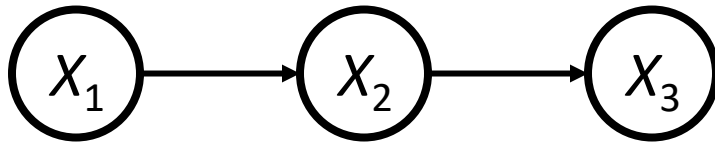
Simple rules for determining whether two variables are dependent or independent.

Identifying independence

Can identify independence relations for any graph with the rules for all Directed Acyclic Graphs of 3 variables (obs and unobs).

E.g., X_1 is sprinklers were on, X_2 is grass is wet, X_3 is it rained...

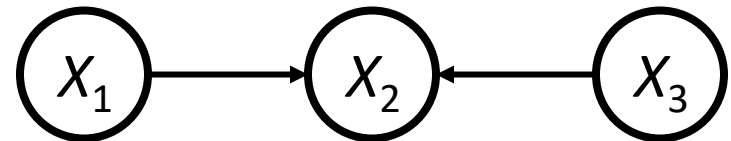
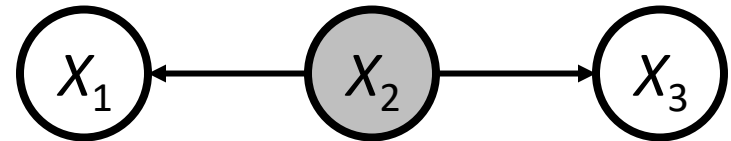
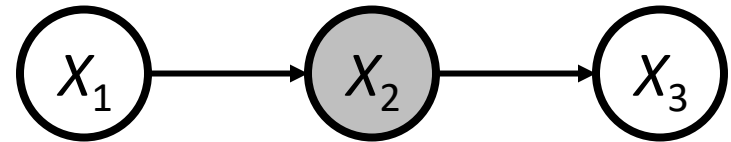
X_1 and X_3 dependent



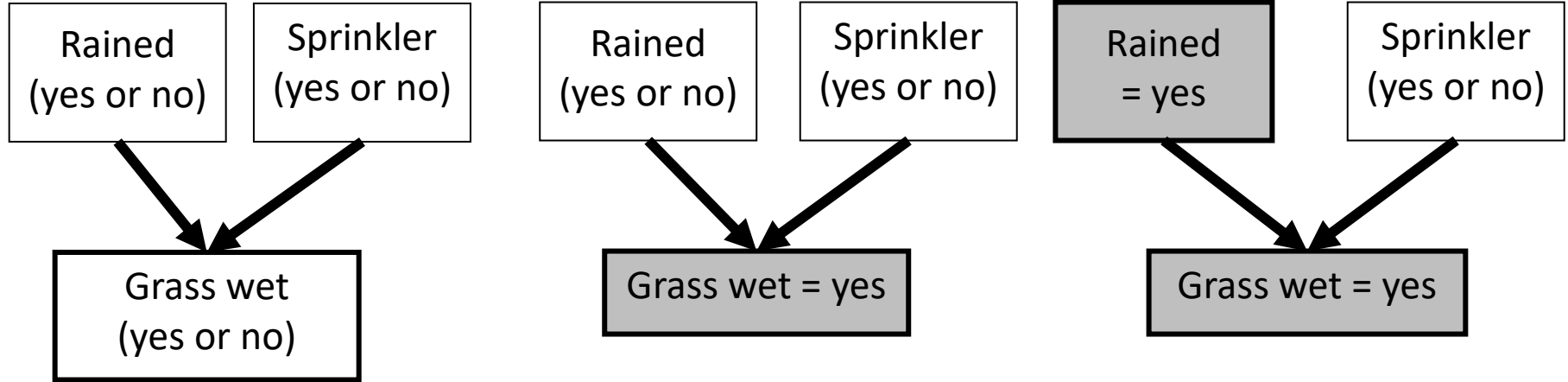
Why?

(shaded variables are observed)

X_1 and X_3 independent



Explaining Away



Corresponding probability relations

Unconditional independence:

$$P(\text{Rained} = \text{yes} \mid \text{Sprinkler} = \text{yes}) = P(\text{Rained} = \text{yes} \mid \text{Sprinkler} = \text{no})$$

After learning effect:

$$P(\text{Rained} = \text{yes} \mid \text{Grass wet} = \text{yes}) > P(\text{Rained} = \text{yes})$$

After learning effect and one cause:

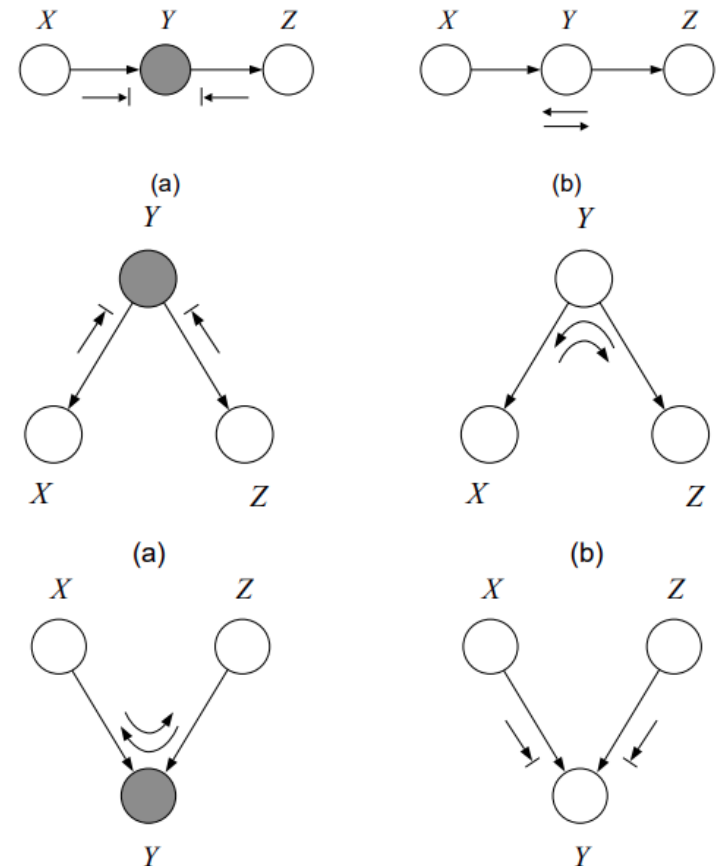
$$\begin{aligned} P(\text{Rained} = \text{yes} \mid \text{Grass wet} = \text{yes}, \text{Sprinkler} = \text{yes}) &= P(\text{Rained} = \text{yes}) \\ &< P(\text{Rained} = \text{yes} \mid \text{Grass wet} = \text{yes}) \end{aligned}$$

Determining (Conditional) Independence: Bayes Ball Algorithm

In more complex Bayesian networks, how do we know which variables are independent of each other?

Bayes-Ball Algorithm:

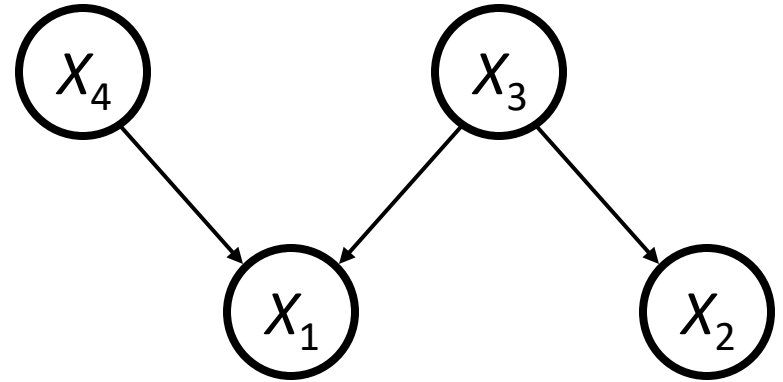
1. Shade conditioned nodes
2. Place balls at one of two nodes of interest
3. Follow rules on right
4. Do balls make it to the other node of interest?



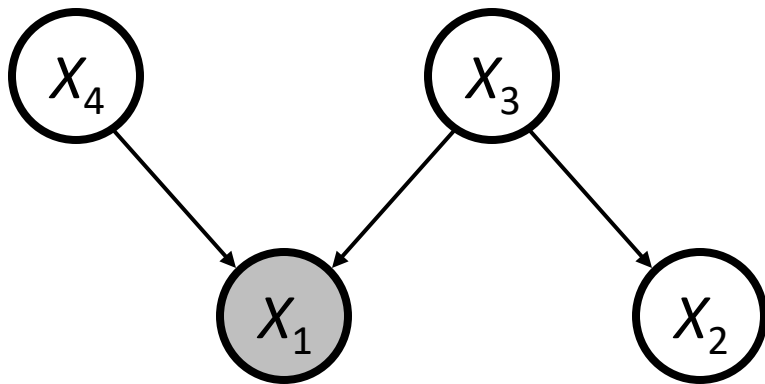
Identifying independence

Four random variables:

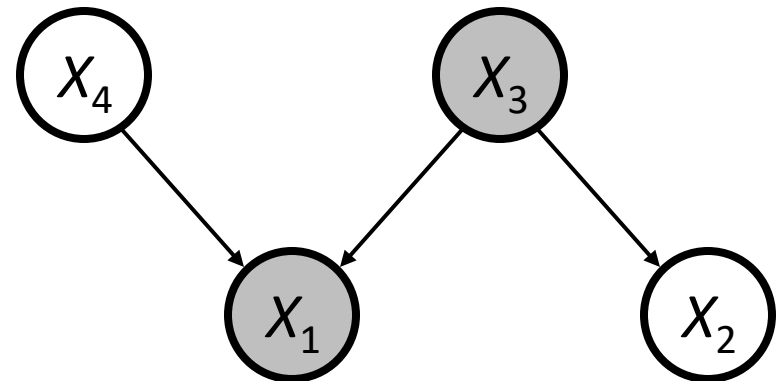
- X_1 coin toss produces heads
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- X_3 friend has psychic powers
- X_4 friend has two-headed coin



X_4 and X_2 are independent



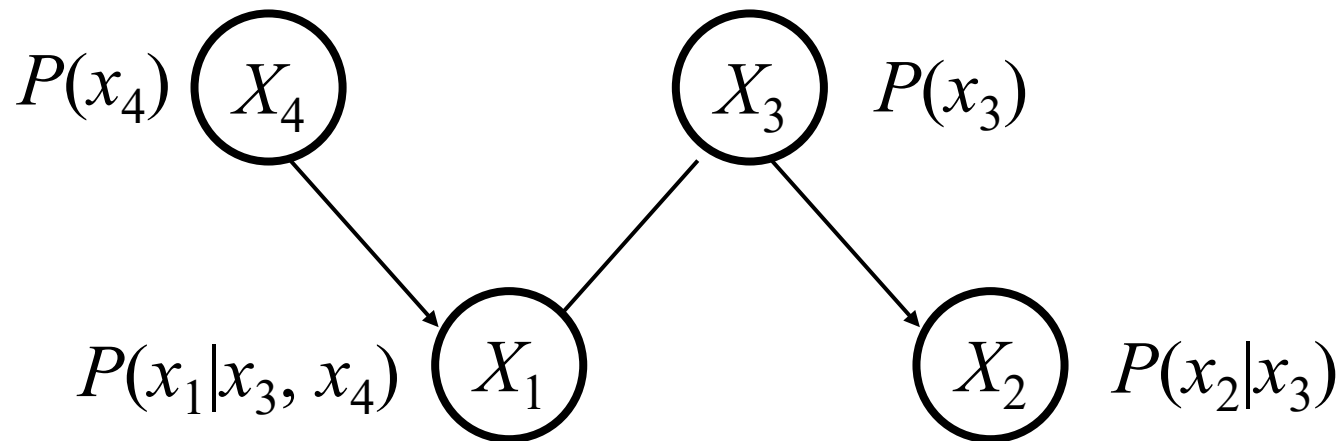
X_4 and X_2 are dependent



X_4 and X_2 are independent

Computing with Bayes nets

$$P(x_2, x_3, x_4 \mid x_1 = 1) = \frac{P(x_1 = 1, x_2, x_3, x_4)}{P(x_1 = 1)}$$

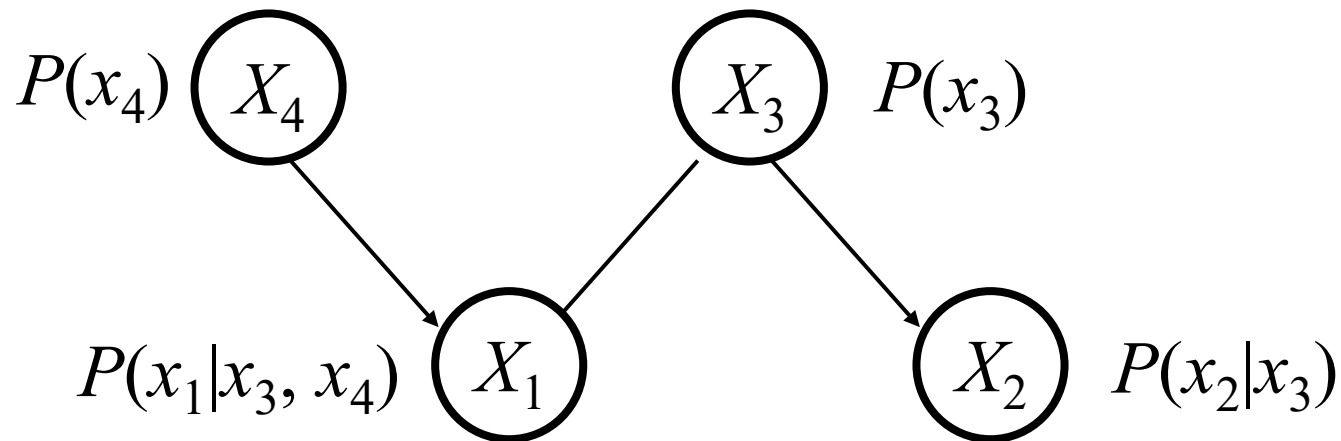


$$P(x_1, x_2, x_3, x_4) = P(x_1|x_3, x_4)P(x_2|x_3)P(x_3)P(x_4)$$

Computing with Bayes nets

$$P(x_1 = 1) = \sum_{x_2, x_3, x_4} P(x_1 = 1, x_2, x_3, x_4)$$

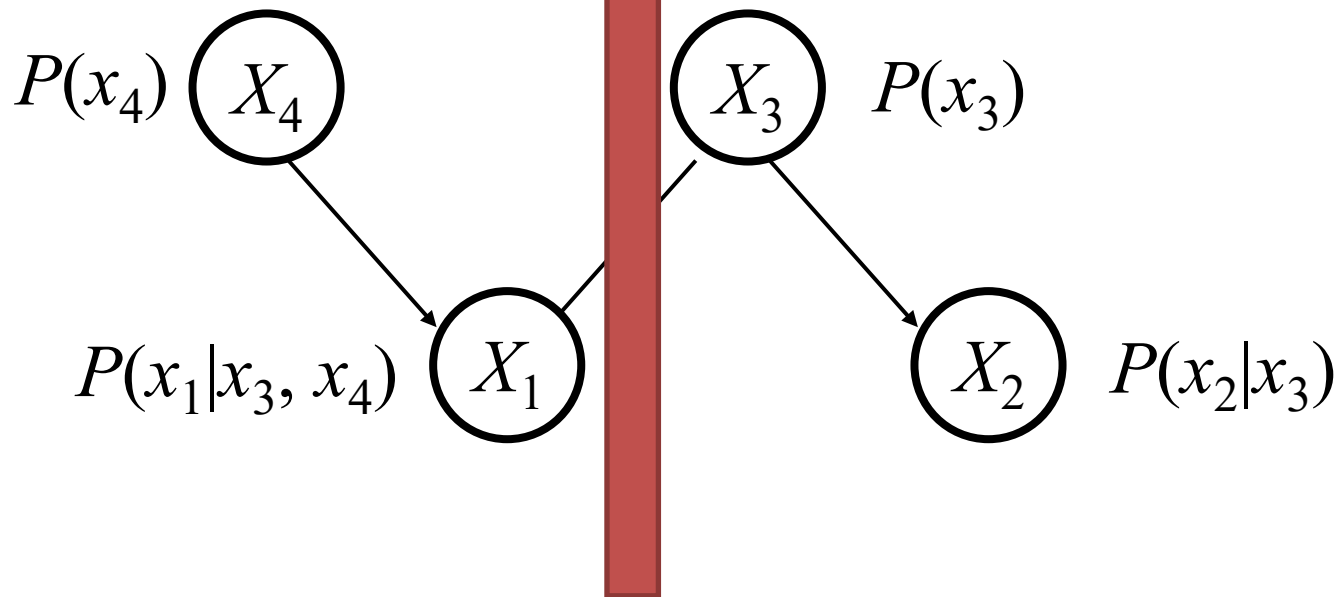
sum over 8 values (values for 2, 3, and 4)



$$P(x_1, x_2, x_3, x_4) = P(x_1|x_3, x_4)P(x_2|x_3)P(x_3)P(x_4)$$

Computing with Bayes nets

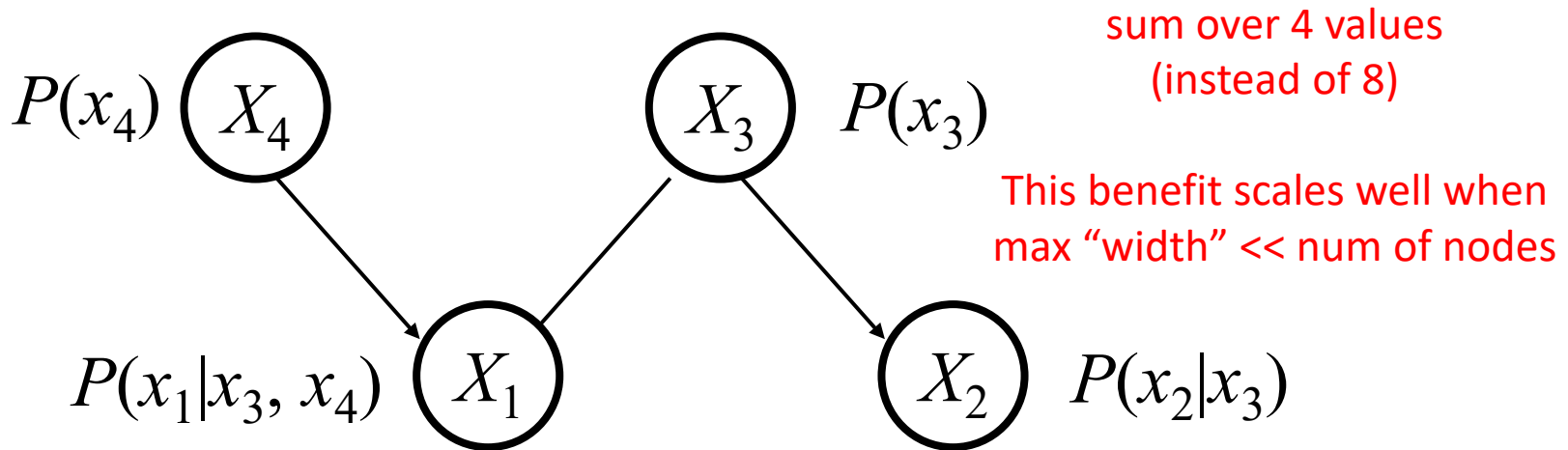
$$P(x_1 = 1) = \sum_{\boxed{x_2}, x_3, x_4}$$



$$P(x_1, x_2, x_3, x_4) = P(x_1|x_3, x_4)P(x_2|x_3)P(x_3)P(x_4)$$

Computing with Bayes nets

$$P(x_1 = 1) = \sum_{x_3, x_4} P(x_1 | x_3, x_4) P(x_3) P(x_4)$$



$$P(x_1, x_2, x_3, x_4) = P(x_1 | x_3, x_4) P(x_2 | x_3) P(x_3) P(x_4)$$

Short rest 1

